Complete Product Catalog — Final

Resolved Questions

#	Question	Answer
1	GTIN format	All GTINs are 14 digits. Excel stripped leading `00`. Pad all with `00` prefix.
2	BWMini serial	Typo corrected: `BWMYYYY-NNNN` (4-digit year), same as BWIII. Start at 3000.
3	BWIII serial	`BWIIIYYYY-NNNN`, 4-digit year. Confirmed.
4	Serial uniqueness	**Global across all BWIII + BWMini products.** BWIII2026-3000 could be any model. One shared sequence.
5	Lot uniqueness	**Global per lot prefix.** A lot like MS60CI-260219 covers all units of that product made that day (could be 600+).
6	Maxxi Position GTIN	Intentionally shared — same product, different compatibility variants. GTIN is the same.

Critical Design Implications

Serial numbers are a SHARED POOL

All 9 serial-mode devices (5 BWIII + 4 BWMini) draw from ONE global serial sequence. The system must:

- Track the **last used serial number** globally (not per product)
- When a user starts a batch for any serial product, suggest `lastUsed + 1` as the start
- Store `data/serial-counter.json` with `{ "lastSerial": 3047 }` (or similar)
- **Prevent duplicate serial assignment** — if BWIII2026-3000 was already printed, it cannot be reused for another product

This changes the architecture: the serial number is NOT tied to a specific product. The `serialStartNumber` in the product profile becomes a **global default starting point** (3000), not a per-product counter.

Two serial prefix families with ONE counter

- BWIII products use prefix `BWIII{YYYY}-`

- BWMini products use prefix `BWM{YYYY}-`
- But they share the same numeric sequence (3000, 3001, 3002...)

Wait — do they really share a counter? If BWIII2026-3000 exists, can BWM2026-3000 also exist? The prefixes are different so the full serial strings are different. **Clarification assumed:** Each prefix family has its OWN counter. BWIII2026-3000 and BWM2026-3000 are different serial numbers because the prefix differs.

→ **System needs TWO counters:** one for `BWIII{YYYY}-` and one for `BWM{YYYY}-`, resetting each year.

Lot numbers = product prefix + manufacturing date

- Lot number is deterministic: `{lotPrefix}-{YYMMDD}`
- One lot covers ALL units manufactured on that date
- E.g., MS60CI-260219 = all Maxxi Gold Snap Electrode 60" units made on Feb 19, 2026
- Batch printing in lot mode = N identical copies of the same label (same lot number)
- No counter needed for lots

Product Summary

Metric	Value
Total products	39
SERIAL mode (with expiration)	9
LOT mode (no expiration)	30
Unique GTIN-14s	37 (Maxxi Position shares 1 GTIN across 3 variants)
Serial prefix families	2 (BWIII, BWM)
Common serial start	3000
Shelf life (serial only)	7 years, all products
Lot formats	30 unique prefixes

SERIAL MODE PRODUCTS (9 devices)

All: 7-year expiration, sequential serials from 3000, NO lot number.

BWIII Family — prefix BWIII{YYYY}-

Serial example: BWIII2026-3000, BWIII2026-3001, ...

Device Name	GTIN-14	Part Number
BWIII Basics	00850008393174	PV2010E
BWIII EEG	00850008393181	PV2006E
BWIII EEG Plus	00850008393198	PV2007E
BWIII PSG	00850008393204	PV2008E
BWIII PSG Plus	00850008393211	PV2009E

BWMini Family — prefix BWM{YYYY}-

Serial example: BWM2026-3000, BWM2026-3001, ...

Device Name	GTIN-14	Part Number
BWMini EEG	00850008393235	PV2310
BWMini HST	00850008393242	PV2312
BWMini HST Compass	00850008393259	SI1842
BWMini PSG	00850008393228	PV2311

LOT MODE PRODUCTS (30 devices)

All: NO expiration, NO serial, lot = {prefix}-{YYMMDD} from manufacturing date.

Maxxi Gold Electrodes (10)

Device Name	GTIN-14	Part #	Lot Prefix
Gold Electrode 48"	00850008393044	PV1010-33CI	MG48CI
Gold Electrode 60"	00867975000295	PV1010-13CI	MG60CI
Gold Electrode 96"	00850008393006	PV1010-23CI	MG96CI
Snap Electrode 48"	00850008393037	SI1872	MS48CI
Snap Electrode 60"	00850008393020	SI1873	MS60CI
Snap Electrode 96"	00850008393013	SI1778B	MS96CI
AgCl 60" Disposable	00850008393303	SI2036	MA60CI
AgCl 60" Reusable	00850008393358	SI2137	MAR60CI
AgCl 96" Disposable	00850008393365	SI2037	MA96CI
AgCl 96" Reusable	00850008393310	SI2138	MAR96CI

Maxxi Flow Sensors (3)

Device Name	GTIN-14	Part #	Lot Prefix
Flow 3ft Key	00850008393341	SI1775B	MF3KCI
Flow 7ft Key	00850008393334	SI1776B	MF7KCI
Flow 7ft TP	00850008393327	SI1486C	MF7SCI

Maxxi Position Sensors (3 — shared GTIN)

Device Name	GTIN-14	Part #	Lot Prefix
Position AC 1.0	00850008393297	SI1247U	MP
Position AC 1.1	00850008393297	SI1247A	MPAC
Position DC	00850008393297	SI1247D	MPDC

Maxxi RIP Respiratory (10)

Device Name	GTIN-14	Part #	Lot Prefix
ABD Interface 3ft	00850008393136	SI2121	IA3
ABD Interface 7ft	00850008393143	SI2124	IA7
Adjustable Belt	00850008393105	SI1659	AB
Infant Belt	00850008393051	SI1660	IB
Large Belt	00850008393075	SI1693	LB
Pediatric Belt	00850008393068	SI1665	PB
THX Interface 3ft	00850008393129	SI2125	IT3
THX Interface 7ft	00850008393112	SI2123	IT7
X-Large Belt	00850008393082	SI1694	XLB
XX-Large Belt	00850008393099	SI1662	XXLB

Maxxi Snore Sensors (3)

Device Name	GTIN-14	Part #	Lot Prefix
Snore 2ft Key	00850008393266	SI1866B	MS2K
Snore 7ft Key	00850008393273	SI1784B	MS7K
Snore 7ft TP	00850008393280	SI1487B	MS7S

Maxxi Cannula (3)

Device Name	GTIN-14	Part #	Lot Prefix
Cannula 7ft N/O	00850008393389	SI1058	MCANO7
Cannula 7ft Nasal	00850008393396	SI1057	MCAN7
Cannula 2ft Nasal	00850008393372	SI1195	MCAN2

Revised Data Schemas

Product (products.json)

```json
{
  "id": "uuid",
  "name": "BWIII PSG Plus",
  "gtin": "00850008393211",
  "partNumber": "PV2009E",
  "description": "",
  "identifierMode": "serial",
  "lotPrefix": null,
  "serialPrefixFamily": "BWIII",
  "shelfLifeYears": 7,
  "shelfLifeMonths": 0,
  "shelfLifeDays": 0,
  "autoExpirationEnabled": true,
  "createdBy": "system",
  "createdAt": "2026-02-18T00:00:00Z",
  "updatedAt": "2026-02-18T00:00:00Z"
}
```

```json
```

```
{
  "id": "uuid",
  "name": "Maxxi Gold Electrode 96\"",
  "gtin": "00850008393006",
  "partNumber": "PV1010-23CI",
  "description": "",
  "identifierMode": "lot",
  "lotPrefix": "MG96CI",
  "serialPrefixFamily": null,
  "shelfLifeYears": 0,
  "shelfLifeMonths": 0,
  "shelfLifeDays": 0,
  "autoExpirationEnabled": false,
  "createdBy": "system",
  "createdAt": "2026-02-18T00:00:00Z",
  "updatedAt": "2026-02-18T00:00:00Z"
}
```

Serial Counter (serial-counters.json) — NEW

```
[
  {
    "id": "bwiii-2026",
    "prefixFamily": "BWIII",
    "year": 2026,
    "lastUsedNumber": 3047,
    "updatedAt": "2026-02-18T10:30:00Z"
  },
  {
    "id": "bwm-2026",
    "prefixFamily": "BWM",
    "year": 2026,
    "lastUsedNumber": 3012,
    "updatedAt": "2026-02-18T10:30:00Z"
  }
]
```

When a batch print job completes:

1. Read the counter for the prefix family + year
2. Update `lastUsedNumber` to the highest serial printed

3. Next batch auto-suggests `lastUsedNumber + 1`

Print Job (print-jobs.json) — serial field update

For serial mode jobs, store the actual serials used:

```json
{
  "serialPrefixFamily": "BWIII",
  "serialYear": 2026,
  "serialRangeStart": 3000,
  "serialRangeEnd": 3049,
  "serialNumbers": ["BWIII2026-3000", "BWIII2026-3001", "..."]
}
```

This creates a complete audit trail of which serial numbers were assigned.

Agent Prompt Revisions Needed

Agent 1 — Phase 1 changes:

1. Add `serial-counters.json` to data storage files
2. Add `serialCounterStore` to stores/index.ts
3. Add `partNumber` and `serialPrefixFamily` to product schema
4. Remove `serialPrefix` and `serialStartNumber` from product (replaced by family)
5. Seed all 39 products on first run
6. Add GTIN-14 note: all GTINs are 14 digits with `00` prefix (Excel strips them)

Agent 1 — Phase 3 changes:

1. Product selector replaces manual GTIN + mode toggle
2. Mode is determined by product selection (not user toggle)
3. Lot number auto-generates from product.lotPrefix + manufacturing date
4. Serial number auto-suggests from serial counter
5. Serial batch builder shows last used number and suggests next range

Agent 2 — Phase 5 changes:

1. HRI text element needs `textAlign` property (left/center/right)
2. ZPL generation uses `^FB` for center/right alignment

Agent 3 — Phase 7 changes:

1. Serial counter update on batch print completion
2. Print job stores serial range for audit trail
3. Duplicate serial prevention check before printing

Agent 4 & 5 — QA changes:

1. Test all 39 products can be selected and generate correct barcodes
2. Test serial counter increments correctly
3. Test lot auto-generation for each prefix
4. Verify GTIN check digits for all 39 products
5. Test shared GTIN (Maxxi Position) works without issues